Sharan Harsoor

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Summary

Principal Machine Learning Engineer with **12+ years** building AI systems at production scale. Inventor on **8 patents** across Citrix and Cohesity in generative AI, intelligent agents, and data security; invention disclosure filed at Red Hat. Author of **two papers submitted to NeurIPS 2026** (under review); international conference speaker (DevConf.IN 2026); creator of open-source AI libraries adopted by startups globally. Deep expertise in agentic AI workflows, RAG pipeline architecture, and LLM deployment for enterprise and security domains. Targeting **Senior Principal Engineer / Senior Staff Engineer** roles in AI/ML, open to **Principal Architect** and **Tech Lead** tracks.

Innovations & Research

Patents – 8 filed across Citrix and Cohesity (USPTO); invention disclosure filed at Red Hat

- Session Failure Prediction in Virtual Applications & Desktop Deployments (**US20230325280A1**, Issued 2023)
- Artificial Intelligence Chatbot for Data Platform Security Analysis (US20250284800A1, Filed Nov 2024)
- Actionable AI Bot for Data Security Correlations (US20250335583A1, Filed Apr 2024)
- Recovery of Compromised Snapshots (US20260064844A1, Filed Aug 2024)
- 4 additional patents in Generative AI and Intelligent Agent domains (pending USPTO)

Publications – NeurIPS 2026 (40th Conference on Neural Information Processing Systems) · Under Review

- *"AmbiguityBench"*: decision-theoretic benchmark, 22,680 trials across 9 frontier models (Google, DeepSeek, Anthropic); novel Ambiguity Premium metric; full open release
- *"Reasoning and Ambiguity Sensitivity in Frontier LLMs"*: cross-provider empirical study on how reasoning modes alter ambiguity-sensitive behavior; implications for safe LLM agent deployment

Conference Talks

- **DevConf.IN 2026, Pune**: "Why Your RAG System Hallucinates: Fixing the Content Segmentation Problem" · **40–60% retrieval accuracy improvement** demonstrated live; open-source: `chunking-strategy` on PyPI
- Talk selected, **DevConf.CZ 2026** (Czech Republic): "Beyond Token Limits: Building Memory That Actually Works for LLM Agents" · topic-aware LLM memory compression; open-source: `llm-smartmem`

Open Source – chunking-strategy (`pip install chunking-strategy`) · llm-smartmem (`pip install llm-smartmem`) · and more

- **chunking-strategy**: Production-grade semantic text chunking with thread-safe parallel processing, streaming pipelines for large files, and adaptive retrieval feedback loops; adopted by multiple startups in production
- **llm-smartmem**: Smart memory management for LLM conversations in agentic production systems

Technical Writing – medium.com/@sharanharsoor · **30+ articles** · **26K+ reads** · 50K+ views

- Topics: RAG architecture, embeddings, LLM observability (LangFuse 3-part series), MCP orchestration, Transformers, Mixture of Experts, production AI system design

Experience

Red Hat — Oct 2025 – Present
Principal Machine Learning Engineer, Data & AI Team — *Bengaluru, India*

- Designed and deployed an **MCP-integrated agentic SKU lookup system** serving **10,000+ Red Hat sales representatives**, enabling intelligent retrieval across **20,000+ product SKUs** filtered by region, country, subscription, and product, eliminating hours of manual lookup per seller weekly
- Architected production RAG pipelines integrating multiple heterogeneous data sources via Model Context Protocol (MCP), forming the retrieval backbone of an enterprise AI assistant serving internal sales operations at scale
- Developed and open-sourced **chunking-strategy** demonstrating **40–60% retrieval accuracy improvement** via semantic-aware segmentation; adopted externally by startups and production engineering teams
- Designed a hash-based incremental indexing system across heterogeneous data sources (GraphQL, web crawlers, databases), cutting daily embedding pipeline cost from **$5,000 to $50** (99% reduction) by eliminating redundant full reindexing of unchanged content
- Filed invention disclosure at Red Hat in intelligent agent and MCP-integrated retrieval domain (pending USPTO filing)
- *Stack: LangChain · LangGraph · Vertex AI · Gemini · pgvector · MCP · React · FastAPI · Python*

Cohesity — Jun 2022 – Sep 2025
Staff Software Engineer, Data Security Squad — *Bengaluru, India*

- Led and delivered the **Cohesity Cyber Recovery Assistant**, the first customer-facing GenAI product in Cohesity's security portfolio, using LangChain's agentic framework; enabled natural language querying of complex cybersecurity data with context-aware security intelligence
- Architected GenAI initiatives: LLM fine-tuning, document embedding pipelines for real-time query response, and multi-agent security workflows; **filed 7 patents** in Generative AI, Security, and Data Science at USPTO

Sr.MTS, Data Science and ML (promoted to Staff SE Oct 2024) *Jun 2022 – Oct 2024*

- Led Anti-Ransomware team of **8 engineers**, designing ML and statistical models across backup adapters; **reduced false positives by 40%** and cut anomaly detection time from **120 to 15 minutes**
- Designed FBProphet-based storage capacity forecasting microservice enabling customers to plan storage needs **up to 30 days in advance**

Citrix Apr 2019 – Jun 2022
Senior Software Engineer II, Performance Analytics Team *Bengaluru, India*

- Reduced system outage detection time from 24 hours to 15 minutes (**93% reduction**) by designing and deploying anomaly detection algorithms for critical customer systems
- Developed time series forecasting and changepoint detection models for Citrix CVAD resources (CPU, RAM), enabling proactive failure prediction and performance capacity planning
- Built ML models for user behavior analysis using MTOP metrics (packet drops, call stability, connection reliability) and **Double Q-learning**, delivering actionable insights for performance management
- Won **Citrix Performance Analytics Hackathon** (ML category) 2020

Cisco May 2015 – Apr 2019
Software Engineer → Software Engineer II *Bengaluru, India*

- Implemented **PAYG licensing** for Cisco ASAv on Microsoft Azure; designed autoscaling feature improving resource utilization on cloud deployments
- Built **"Jeeves"** (Cisco ThingQbator innovation program), a cloud-based NLP system for natural language instruction execution; developed ML regression POC for hardware health monitoring on Cisco ASR-9000
- Finalist, **Intel India Embedded Challenge (IIEC)**, Professional Category 2014 & Student Category 2012

Education

University of Arizona 2021 – 2023
Master of Science, Data Science | ***CGPA: 3.889 / 4.00*** *Tucson, AZ, USA*

- Perfect 4.0 in: Machine Learning, Neural Networks, Applied NLP, Data Mining, Cloud Analytics, Data Warehousing & Analytics

B.M.S. College of Engineering, Bangalore 2009 – 2013
Bachelor of Engineering, Electronics & Communication | ***CGPA: 8.44 / 10***

Indian Institute of Science (IISc) 2021 – 2022
Reinforcement and Deep Reinforcement Learning, CCE Certification

Skills

Agentic AI & LLMs: LangChain, LangGraph, LlamaIndex, HuggingFace, OpenAI API, Vertex AI, Gemini, Claude, RAG pipelines, MCP (Model Context Protocol), agentic workflows, prompt engineering, LLM fine-tuning, RLHF, LangFuse

ML / DL: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, FBProphet, NumPy, Pandas, anomaly detection, time series forecasting, changepoint detection, statistical modeling, reinforcement learning, Double Q-learning, NLP, computer vision

Security AI: ransomware detection, cybersecurity AI, data security, anomaly detection systems, cyber recovery, threat intelligence

Vector & Retrieval: pgvector, Chroma, Pinecone, semantic chunking, cross-encoder re-ranking, embeddings, vector search

Data Engineering & ETL: Apache Spark, PySpark, Apache Kafka, ETL pipelines, Azure Data Factory, data pipelines, data warehousing, distributed systems

MLOps & Infra: FastAPI, Docker, Kubernetes, MLflow, microservices, REST APIs, CI/CD

Cloud: Google Cloud (Vertex AI), Microsoft Azure, AWS

Languages: Python, Go (Golang), C++, C, SQL, JavaScript/React, Bash/Shell

Certifications

Generative AI with LLMs (DeepLearning.AI) · LangChain & Vector Databases in Production (Activeloop) · Reinforcement & Deep RL (IISc) · Deep Learning (NPTEL IIT Kharagpur) · Deep Learning (PadhAI) · Apache Kafka (Conduktor) · PySpark (Udemy) · FastAPI Masterclass (Udemy)

Awards & Recognitions

Winner, Most Innovative Project, Cohesity GenAI Hackathon 2023 · Winner, Citrix Performance Analytics Hackathon (ML) 2020 · Kaggle Master

Languages

English (Full professional) · Kannada (Native) · Hindi (Native) · German (Elementary)